NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Unusual Does Not Mean Impossible

AP Statistics · Unit 2 · Topic 2.10 · B: 2026-11-06 · E: 2026-11-30

[STAR] TODAY'S PROBLEM

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: "Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies."

Analyst: "A result can be unusual without being impossible; compare 58 with the model's mean and standard deviation."

The binomial model gives a chance of about 0.0072 for 58 or fewer makes; use this provided value.

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE — before the video (5 min)

Whose statement is more defensible, the coach's or the analyst's? Name one quantity you would check.

[BOOK] CENTER, SPREAD, AND UNUSUAL COUNTS (keep on the page)

For a binomial count, $\mu_X = np$ and $\sigma_X = \sqrt{np(1-p)}$, both in count units. The mean is an average over many comparable sets of trials. The standard deviation describes a typical distance from that mean. A result far from the mean may be unusual, but a small positive probability does not mean impossible or prove that the model changed.

Finish after the video:

DOK 1 **(a)** State n and p for this model.

DOK 2 **(b)** Calculate the binomial mean and standard deviation. Interpret the mean for many sets of 100 shots.

DOK 3 **(c)** Whose statement is supported? Compare 58 with the mean and standard deviation, and use the given probability to distinguish unusual from impossible. Explain what the coach's claim could make a reader wrongly believe. Write 3–4 sentences.

Frame: The result 58 is ________ makes below the mean, compared with an SD of about ________ makes.

Frame: A chance of 0.0072 means ____________________, not ______________, so the coach's claim could mislead by ________________________.

EXIT — turn this in

One thing the video changed about my first take: ____